Supplementary Material: Directional Prompting for Gaze-Referential Inference in Large Multimodal Models

Anonymous submission

Anonymous institution

This document accompanies the MMM 2027 submission. It collects dataset-construction details, prompt templates, additional tables, qualitative examples, and implementation hyperparameters. The main paper is self-contained; this file is for reproducibility.

1 Construction of CG-GRI

CG-GRI (N=483) is derived from VideoAttentionTarget [1] (N=200) and GazeFollow [3] (N=283). Neither source provides GRI-ready candidate lists, so we construct MCQ options and targets as follows. We write *VAT-GRI* and *GF-GRI* for these two subsets of CG-GRI; they are reporting slices of a single track, not additional evaluation tracks.

VideoAttentionTarget (VAT-GRI). VideoAttentionTarget consists of video clips with dense per-frame gaze-target annotations. Consecutive frames within a clip are visually similar, so we sample by gaze sequence / clip and keep at most one frame per clip for the frozen experimental release. Automatic screening uses YOLOv10L [4] detections: the observer head box must be valid; detections overlapping the observer box are removed; the annotated gaze point must fall in at least one remaining box; and at least two candidate boxes remain. Short natural-language descriptions of the observer and each candidate are then written (LMM draft + human edit) so that instance phrases are visually distinguishable (e.g. two cups described by appearance or relative position, not by box coordinates). Every retained row is human-reviewed: observer identity, candidate list, and the target implied by the original VAT gaze point. Ambiguous or unverifiable images are discarded. Ground-truth arrow origins are human-confirmed eye points: each UniGaze-predicted eye start is reviewed manually and either accepted or replaced by a manually clicked correction; endpoints remain the original VAT gaze points. The ground-truth direction is therefore the ray from the confirmed eye point to the annotated gaze point. This yields the frozen **200** VAT-GRI instances used in the main paper.

GazeFollow (GF-GRI). GazeFollow provides web images annotated with gazed-region heatmaps but not with gazed-object labels. We randomly sample 1,000 images and retain, by manual judgment, only those in which the observer's eyes are clearly visible. We discard images in which the observer looks away from

the camera, the gaze direction is occluded, or eye visibility is unclear. For each retained image we (1) list candidate objects (YOLO draft + human merge/add) and (2) label the ground-truth gazed target from the heatmap. Ground-truth arrows use the original GazeFollow eye annotation as the origin and the original gaze point as the endpoint. This yields **283** GF-GRI instances.

Shared protocol. On CG-GRI the observer is marked with a red bounding box to disambiguate multi-person scenes; the box is *not* used as the arrow origin. It serves two other purposes. It is drawn in every CG-GRI condition, including the RGB baseline, so that the only difference between conditions is the directional cue. It also selects whose gaze to predict: among the faces detected in the full image, we take the largest one overlapping the box as a reference head, run the gaze estimator once per overlapping face with a tight box around it, and keep the run whose projected origin lands closest to that reference head. The origin itself always comes from the estimator, never from the box. FG-GRI does not use a head box: it uses the original candidate labels and face-centered arrow origins. The multiple-choice question lists the verified candidate names and asks which object the observer is looking at. Ground-truth and UniGaze arrows (and matched textual coordinates) are built from the same arrow record.

2 Prompt Templates

All GRI conditions share the same zero-shot MCQ stem and JSON parser unless the condition *is* a prompt change (textual coordinates, CoT, CoGT, or LMM-generated Stage 1). Answers are parsed from the JSON `answer` field and scored by case-insensitive exact string match against the ground-truth label. `{choices}` is replaced at runtime with candidate names, one per line.

FG-GRI (plain). Used for RGB, D, RGB+D, RGB+A, D+A, RGB+D+A, and Stage 2 of the LMM-generated-arrow protocol (the overlay already contains the arrow).

Prompt (FG-GRI)

```
What is the person looking at in the image? You must choose
exactly one option from the following options:
{choices}
Please answer in the following format:
```json{"answer": "<your chosen option>", "reasoning": "<your
reasoning>"}```
```

CG-GRI (plain). Same stem, but the observer is identified by the red bounding box.
```

Prompt (CG-GRI)

```
What is the person in the red bounding box looking at?
You must choose one from the following options:
{choices}
Please answer in the following format:
```json{"answer": "<your chosen option>", "reasoning": "<your
reasoning>"}```
```

*Textual-arrow prompting (*RGB+A$_{\text{txt}}$*).* The image remains clean RGB (plus the observer box on CG-GRI). The same start–end geometry as the gaze-direction overlay is inserted after the question and before the choice list. Coordinates are ground-truth for RGB+A$_{\text{txt}}$(GT) or UniGaze-predicted for RGB+A$_{\text{txt}}$(UniGaze). On CG-GRI the first sentence uses "the person in the red bounding box" as above.

Prompt with textual arrow (RGB+A$_{\text{txt}}$, FG-GRI wording)

```
What is the person looking at in the image? The target person's
gaze ray originates from pixel ({x1}, {y1}) and ends at pixel ({x2},
{y2}). The origin (0, 0) is at the top-left corner of the image.
You must choose exactly one option from the following options:
{choices}
Please answer in the following format:
```json{"answer": "<your chosen option>", "reasoning": "<your
reasoning>"}```
```

CoT and CoGT (FG-GRI). The exact common MCQ stem and the CoT and CoGT suffixes are reported in the main paper. CoGT uses RGB only and is not combined with RGB+A.

LMM-generated arrow, Stage 1. Stage 1 sees RGB only (plus the observer box on CG-GRI), *without* answer choices, and must emit a gaze arrow. A deterministic renderer overlays that arrow; Stage 2 then uses the plain MCQ prompt above. Malformed or implausibly short Stage 1 arrows are recorded as failures and are never replaced by a UniGaze or ground-truth arrow. On CG-GRI the minimum usable arrow length is 35 px. Both LMMs produce a valid arrow for every FG-GRI image. On CG-GRI, Gemini-3.1-pro fails once (one GazeFollow arrow below the length floor) and Qwen3.5-27B fails 15 times (one on VideoAttentionTarget and 14 on GazeFollow, all below the floor); these are scored incorrect. A further two GazeFollow images were declined by the Qwen API, which is why its CG-GRI total is 481 rather than 483.

LMM-generated Stage 1 (FG-GRI)

```
Identify the target person's gaze direction in this image. Do not
name an object.
Return exactly one JSON object, using a normalized 0-1000 image
coordinate system: top-left is (0,0), bottom-right is (1000,1000).
The ray start must be near the target person's visible eyes. The ray
end must be where that person is looking.
Format: {"start": [x, y], "end": [x, y]}
All four values must be numbers between 0 and 1000 inclusive.
```

LMM-generated Stage 1 (CG-GRI)

```
The person whose gaze you must estimate is marked by a red bounding
box.
Infer that person's gaze direction from the image. Do not identify an
object and do not list any answer choices.
Return exactly one JSON object and nothing else. Use a coordinate
system in which the top-left is (0, 0) and the bottom-right is (1000,
1000):
{"start": [x, y], "end": [x, y]}
Both points must be within 0 through 1000. The start must be near the
observer's eyes and the end must lie in the gaze direction.
```

Point / line controls. Endpoint-only (point) and headless-shaft (line) conditions use the same plain MCQ prompt as RGB+A; only the overlay rendering changes.

3 Predicted-Arrow Construction and Endpoint Rule

From a 3D gaze direction to a 2D arrow. UniGaze returns a 3D gaze vector, which we project into the image before any depth reasoning. Six facial landmarks—the four eye corners and the two nose corners—are fitted to a 3D face model by `solvePnP` under a fixed pinhole camera (focal length $4W$ for image width W, principal point at the image center, no distortion), which gives the head rotation and translation. The arrow origin is the projection of the resulting 3D face center, the mean of those six landmarks, which is also the point UniGaze regresses its direction from. Projecting a second 3D point at a fixed distance along the predicted vector gives a hint pixel that fixes only the 2D *direction* of the arrow; its length comes from the depth map, so the choice of that distance does not affect the rendered cue.

Terminating the ray on the depth map. The 2D ray is extended from the origin through the hint pixel to the image border, whose intersection (u_e, v_e) serves as a reference. Writing z_0 and z_e for the Depth Anything V2 values at the origin and at (u_e, v_e), we sample the ray every 0.75 px between 10 px (skipping the observer's own face) and 800 px, and compare the scene depth $z_{\text{scene}}(t)$ with a ray depth interpolated linearly in image space, $z_{\text{ray}}(t) = z_0 + \frac{t}{\|(u_e,v_e)-(u_0,v_0)\|}(z_e - z_0)$.

Depth Anything V2 predicts relative inverse depth, so larger values are nearer and the ray is blocked wherever $z_{\text{ray}}(t) \leq z_{\text{scene}}(t) + \tau$, with the tolerance τ set to 2% of the image's valid depth range (1st–99th percentile). The first samples of the maximal contiguous intervals satisfying this test are the first and second occluders; the border intersection is used when no interval exists. The test is deliberately relative: it needs neither metric depth nor camera calibration beyond the fixed pinhole model above.

Ground-truth arrows. Ground-truth visual and textual arrows are manually prepared for every retained instance on both tracks. On FG-GRI, the annotated gazed point defines the arrow tip; on CG-GRI, the annotated gaze target defines the endpoint. Construction details for each source dataset are given below.

Predicted arrows. Predicted arrows use the same rendering protocol but take their endpoints from UniGaze direction and first-occluder depth grounding. The A(GT) versus A(UniGaze) comparison therefore isolates endpoint error rather than a change of rendering protocol. FG-GRI is built from the pilot stimuli of Zhang et al. [6]; their public release contains 907 images, whereas the paper describes the pilot pool as 900.

Which occluder to use. A UniGaze-predicted arrow is terminated on the monocular depth map at the *first occluder* (physical default: first intersection with estimated scene geometry), the *second occluder*, or the *image boundary*. The comparison is a sensitivity analysis of that default, not a test-set choice of method. On FG-GRI (N=904), Gemini-3.1-pro and Qwen3.5-27B rank the three rules the same: first occluder (86.4% / 72.2%) ahead of the second occluder (82.2% / 69.0%) and the image boundary (80.1% / 62.4%). For Gemini the pairwise drops are Δ=−4.2 pp (p=0.0018) and Δ=−6.3 pp ($p<10^{-5}$); both models' pairwise differences from the default are significant (paired McNemar, $p<0.05$). Even boundary termination remains well above each model's RGB baseline (55.5% / 44.6%). One reason first-occluder ranks highest is specific to FG-GRI: tabletop scenes place the labeled target on the nearest surface along the ray, so the first occluder usually coincides with the annotated endpoint. Reported UniGaze RGB+A numbers keep this physically motivated default; we do not treat it as universally optimal.

4 Arrow Length and Direction Sensitivity

We further test whether FG-GRI accuracy is sensitive to errors in the *geometry* of a ground-truth visual arrow, isolating cue quality from UniGaze prediction noise. On Gemini-3.1-pro and Qwen3.5-27B, we shorten the arrow to α of its original length with $\alpha \in \{1, \frac{2}{3}, \frac{1}{3}\}$, or rotate it by $\Delta\theta \in \{-30, -20, -10, 0, 10, 20, 30\}$ degrees about the eye while keeping the target label unchanged. Figure 1 shows that accuracy falls monotonically as the arrow shortens and as rotation increases in magnitude; every perturbed condition differs significantly from the unperturbed

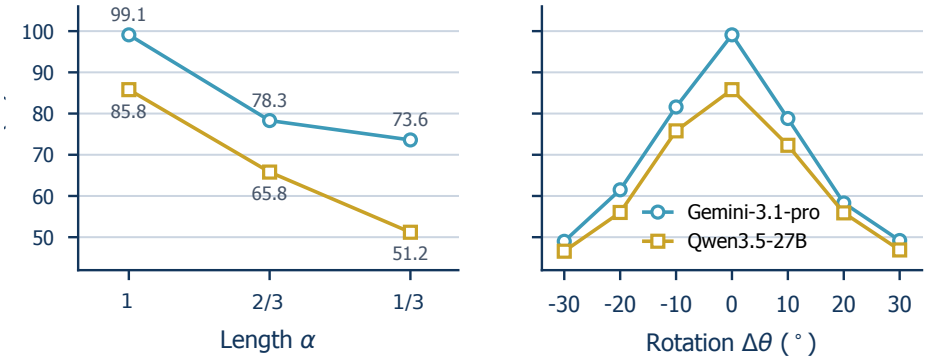


Fig. 1. Sensitivity of FG-GRI accuracy to ground-truth arrow length and direction for `Gemini-3.1-pro` and `Qwen3.5-27B` under RGB+A(GT).

arrow (paired McNemar, all $p < 0.01$). The pattern is therefore a property of the cue rather than of one LMM.

5 Why Does Gemini-3.1-pro Drop in the Three-Candidate Stratum?

The main-paper stratified table shows an unusual CG-GRI result for Gemini-3.1-pro: the changes from RGB to RGB+A(UniGaze) are +2.5, −8.0, and +2.9 pp for 2, 3, and 4 candidates, respectively, and only the 3-candidate decrease is significant (McNemar $p<0.05$). This decrease appears independently in VAT-GRI (−7.5 pp, $N=79$) and GF-GRI (−8.4 pp, $N=107$), so it is not explained by one source alone. However, the candidate-count strata are not matched, controlled subsets and differ in size and composition; the isolated decrease should therefore not be interpreted as a causal effect of having exactly three candidates.

A paired diagnostic indicates that Gemini is especially responsive to the directional cue. The predicted arrow overturns 63 of its 380 correct RGB decisions (16.6%) while correcting 56 of its 103 RGB errors (54.4%); the corresponding rates for GPT-5.2 are 10.1% and 42.0%. Qualitative inspection of the overturned cases found examples in which a misaligned predicted arrow redirected Gemini to an incorrect candidate. By contrast, a ground-truth arrow raises Gemini's overall CG-GRI accuracy from 78.7% to 95.0%, ruling out a generic overlay penalty. These post-hoc results suggest that inaccurate predicted arrows can override Gemini's otherwise correct RGB-based decisions, while accurate arrows remain highly beneficial.

Table 1. CG-GRI accuracy under UniGaze and LMM-generated arrows (main-paper spatial-cue table, CG-GRI rows). Qwen3.5-27B uses N=481; other models N=483.

Model	RGB+A(UniGaze)	RGB+LMM-gen	Δ	McNemar p
Qwen3.5-27B	77.5	72.0	−5.5	0.012
GPT-5.2	76.2	72.0	−4.1	0.054
Gemini-3.1-pro	77.2	81.8	+4.6	0.068

6 LMM-Generated Arrow: Gaze Alignment on CG-GRI

The main paper reports RGB+LMM-gen alongside RGB+A(UniGaze) in the spatial-cue comparison table (CG-GRI rows; Qwen3.5-27B N=481, others N=483). Table 1 reproduces those accuracies. On CG-GRI the sign differs by model: Gemini-3.1-pro reaches 81.8%, above its 77.2% UniGaze accuracy (+4.6 pp over paired instances; McNemar p=0.068), whereas Qwen3.5-27B and GPT-5.2 fall below their UniGaze results. This section documents a post-hoc audit that asks whether those CG-GRI differences co-vary with how well each overlay aligns with the ground-truth gaze ray. It is exploratory and does not change any main-paper numbers.

Metrics and evaluable subset. For each instance we compute (i) *angular error*, the angle between the overlay ray and the ground-truth gaze direction, and (ii) *endpoint distance*, the Euclidean distance from the overlay endpoint to the annotated ground-truth gaze point (pixels). We report medians over the *geometry-evaluable* CG-GRI subset (N≈389–405 per model): Stage-1 must succeed, and on GazeFollow every multiple-choice option needs a YOLO box match so that gaze geometry can be scored. Accuracies in Table 1 remain the full CG-GRI totals from the main paper (N=481 for Qwen3.5-27B, 483 otherwise).

Coordinate conversion. LMM-generated Stage-1 rays are emitted on a normalized 0–1000 grid (Section 2). Before any distance or angle computation, self-generated endpoints are mapped to pixel space as $x_{\text{px}} = x_{\text{norm}}(W-1)/1000$ (and analogously for y), matching the renderer in `run_vat_self_generated_arrow.py`. GazeFollow ground-truth and UniGaze coordinates are stored in $[0, 1]$ and scaled by image width and height; VideoAttentionTarget gaze points are already in pixels. All comparisons use the ground-truth *gaze point*, not candidate bounding boxes.

Pooled alignment. Table 2 shows that Gemini self-generated endpoints are median-closer to the ground-truth gaze point than UniGaze overlays on this subset (36 px vs. 152 px) and carry smaller median angular error (4.8° vs. 12.3°). GPT-5.2 and Qwen3.5-27B do not show the same endpoint advantage; GPT-5.2 self endpoints are in fact farther from the gaze point (188 px). Table 3 splits Gemini by source: on both VAT-GRI and GF-GRI, self-generated endpoints are closer to the annotated gaze point than UniGaze (VAT: 46 vs. 286 px; GF: 28 vs. 95 px).

Table 2. Median gaze alignment on the geometry-evaluable CG-GRI subset. UniGaze geometry is shared across models; accuracies are per-model full-CG totals from Table 1.

Cue / model	Angle (°)	Endpoint dist. (px)	Acc. (%)	n_{geom}
UniGaze	12.3	152	77.5 / 76.2 / 77.2 (Qwen/GPT/Gemini)	≈400
Gemini LMM-gen	4.8	36	81.8	405
GPT-5.2 LMM-gen	10.4	188	72.0	400
Qwen3.5-27B LMM-gen	9.1	98	72.0	389

Table 3. Gemini-3.1-pro gaze alignment by CG-GRI source (geometry-evaluable instances; medians).

Source	n	UniGaze dist. (px)	Self dist. (px)	UniGaze angle (°)	Self angle (°)
VAT-GRI	200	286	46	11.4	5.4
GF-GRI	205	95	28	12.9	4.2

Instance transitions RGB+A(UniGaze)→RGB+LMM-gen. To relate accuracy changes to cue quality, we classify each paired instance by whether UniGaze and the self-generated overlay agree: *fix* (UniGaze wrong, self correct), *break* (UniGaze correct, self wrong), *stable_ correct*, or *stable_ wrong*. On full CG-GRI (N=483), Gemini-3.1-pro has 77 fixes and 55 breaks under this definition, consistent with its net +4.6 pp gain. Table 4 reports median geometry on the geometry-evaluable subset for Gemini. *Fix* cases combine better self alignment (median 5.4° angular error, 28 px endpoint distance) with UniGaze errors (15.3°, 174 px). *Break* cases reverse the pattern: UniGaze remains well aligned (8.4°, 192 px) while self-generated direction error explodes (median 66.3°), yielding 0% self accuracy by construction. Gemini's CG-GRI gain is therefore fix-heavy and consistent with better self-drawn gaze geometry on instances where UniGaze fails, not with uniformly better pooled endpoint placement alone.

Angular-error bins (exploratory). On the geometry-evaluable subset, Gemini self accuracy rises to 95.7% when self angular error is $\leq 10°$ (n=208) but falls to 49.4% when error exceeds 30° (n=85), whereas UniGaze accuracy varies much less across the same bins (82.3% vs. 73.6%). Endpoint distance correlates negatively with self accuracy for Gemini (Spearman $\rho \approx -0.25$, $p<0.001$ on this subset).

Reproducibility. The audit is generated by `gaze_stimuli/analysis/cg_gri/analyze_special_models.py` (command: `python analysis/cg_gri/analyze_special_models.py` from `gaze_stimuli/`). Per-instance outputs are written to `gaze_stimuli/analysis/cg_gri/special_models/`. A markdown working note with the same numbers is kept at `docs/CG_GRI_LMM_GEN_GAZE_ALIGNMENT_AUDIT.md`

Table 4. Gemini-3.1-pro: median gaze alignment by RGB+A(UniGaze)→RGB+LMM-gen transition type (geometry-evaluable CG-GRI). Acc. is self (LMM-gen) accuracy within each row.

Transition	n	Self angle (°)	UniGaze angle (°)	Self dist. (px)	UniGaze dist. (px)	Self acc. (%)
fix	61	5.4	15.3	28	174	100
break	45	66.3	8.4	336	192	0
stable_correct	277	3.5	12.2	26	131	100
stable_wrong	22	18.3	9.3	391	237	0

7 Stratified Error-Type Distribution

The main paper reports the error-type definitions, representative examples, and aggregate counts. Table 5 further stratifies the same 123 failures by view, proximity, and number of candidates. Inaccurate cues dominate almost every bucket; geometric ambiguity, reasoning hallucination, and depth-order confusion remain rare. Right-view errors show a higher share of reasoning hallucination (18.5%) than left-view errors (3.0%).

Table 5. Stratified error-type distribution (`Gemini-3.1-pro`, RGB+A(UniGaze), FG-GRI; N=123 failures). Each cell: count (%).

Stratification	(a) Inaccurate Cue	(b) Geometric Ambiguity	(c) Reasoning Hallucination	(d) Depth-order Confusion
By View Angle				
Left (N=66)	59 (89.4%)	2 (3.0%)	2 (3.0%)	3 (4.5%)
Right (N=54)	37 (68.5%)	4 (7.4%)	10 (18.5%)	3 (5.6%)
Front (N=3)	3 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
By Proximity				
Level 1 (closest, N=68)	52 (76.5%)	6 (8.8%)	8 (11.8%)	2 (2.9%)
Level 2 (medium, N=30)	24 (80.0%)	0 (0.0%)	4 (13.3%)	2 (6.7%)
Level 3 (farthest, N=25)	23 (92.0%)	0 (0.0%)	0 (0.0%)	2 (8.0%)
By Number of Options				
2 options (N=60)	45 (75.0%)	5 (8.3%)	7 (11.7%)	3 (5.0%)
3 options (N=34)	26 (76.5%)	1 (2.9%)	4 (11.8%)	3 (8.8%)
4 options (N=29)	28 (96.6%)	0 (0.0%)	1 (3.4%)	0 (0.0%)

8 Limitations of Depth as an LMM Input

Adding a depth-color arrow on a depth plate (D+A) yields limited and inconsistent gains relative to adding a gaze-direction arrow on RGB (main-paper overall table). On FG-GRI, Gemma-4-26B improves only +3.8 pp from D to D+A(GT) (39.9%→43.7%), far below +41.7 pp from RGB to RGB+A(GT) (50.8%→92.5%). Candidate objects and the pseudo-color shaft are harder to recognize on depth

plates (main-paper Figure 1); some LMMs describe depth images as thermal or infrared scenes.

An occasional D+A(UniGaze)>D+A(GT) gap on depth plates should not be read as UniGaze outperforming manual gaze in general: on RGB+A, GT arrows dominate for every model. On FG-GRI D+A, UniGaze endpoints exceed GT accuracy for several LMMs (largest for Gemma-4-26B, Δ=+7.2 pp, McNemar $p<0.001$), which we attribute to colormap / first-occluder artifacts on brittle depth rationales rather than a better 3D gaze estimate.

9 CoT Failure Examples

Generic CoT does not reliably improve GRI when a ground-truth gaze-direction arrow is already present (main-paper CoT table). We inspect RGB+A(GT)+CoT failures of Gemini-3.1-pro on FG-GRI: the image contains the ground-truth gaze arrow plus the CoT suffix. CoT can still misread or override the arrow.

*Example 1 (ground truth: **coffee**).* RGB+A(GT)+CoT returns **doll**, stating that the gaze arrow "points directly at the pink doll" even though the arrow points at the cup of **coffee**. The doll and coffee are proximity level 1 (nearest).

*Example 2 (ground truth: **remote control**).* RGB+A(GT)+CoT returns **water**, stating that the gaze arrow "points directly at the bottle of water" even though the arrow points at the **remote control**. The shaft passes near the bottle while both objects are close.

When candidates are near one another, step-by-step text can hallucinate a target that contradicts the overlay.

10 Implementation Details

Depth Anything V2 [5]. ViT-Large (*vitl*) encoder, input size 518. Relative depth is normalized per image to $[0, 1]$ using the 1st and 99th percentiles of valid values, then rendered for depth-colored arrows with a viridis colormap.

UniGaze [2]. Input resized to 448×448. We take the gaze direction from the projected 3D face center and find its first intersection with scene geometry in the depth map (first occluder; fallback: image boundary); see Section 3.

YOLOv10L [4]. Confidence threshold 0.25, NMS IoU 0.45 (Ultralytics defaults). Heuristic baseline: exclude "dining table" on all tracks and "person" on FG-GRI only (never a FG-GRI ground-truth answer).

Heuristic baseline scoring. The heuristic baseline selects the candidate whose YOLOv10L bounding-box center is nearest to the predicted endpoint. We score it by manual review—one author inspects each selected box and marks the instance correct if the box covers the ground-truth gaze target. Because the correctness criterion (box coverage) differs from the LMM protocol (one named candidate, exact-match parsing), this accuracy is a rough reference point for how far a purely geometric pipeline gets, not a strictly comparable number.

LMM decoding. Proprietary models were queried via commercial APIs in April–May 2026 using the identifiers `gpt-4.1`, `gpt-5.1`, `gpt-5.2`, `gemini-2.5-flash`, and `gemini-3.1-pro-preview`. Decoding uses temperature 0 wherever the provider accepts it, falling back to the provider default only for models that reject the parameter; each instance is queried once per condition. Answers are recovered by a JSON-constrained response or a post-hoc parse of the `answer` field, depending on provider support. Open-source models use the same prompt templates and parser.

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